Clément Dumas

Astra Fellow with Owain Evans, previously MATS 7 scholar with Neel Nanda.

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PUBLICATIONS

| lead-author papers ✦ main contributors

Training with conflicting values can induce CoT override
C. Dumas✦, J. Treutlein, J. Betley, O. Evans
LessWrong 2026

Cooperation with AIs seems to be a low-hanging fruit for better eval practices
C. Dumas✦
LessWrong 2026

Steering towards "automated grading" degrades alignment
J. Betley✦, J. Treutlein, **C. Dumas**
LessWrong 2026

Narrow Finetuning Leaves Clearly Readable Traces in Activation Differences
J. Minder✦, **C. Dumas**, S. Slocum, H. Casademunt, C. Holmes, R. West, N. Nanda
ICLR 2026; NeurIPS 2025 Mechanistic Interpretability Workshop (Spotlight)

Overcoming Sparsity Artifacts in Crosscoders to Interpret Chat-Tuning
J. Minder✦, **C. Dumas**✦, C. Juang, B. Chughtai, N. Nanda
NeurIPS 2025

What We Learned Trying to Diff Base and Chat Models (And Why It Matters)
C. Dumas✦, J. Minder✦, N. Nanda
Alignment Forum 2025

Separating Tongue from Thought: Activation Patching Reveals Language-Agnostic Concept Representations in Transformers
C. Dumas✦, C. Wendler✦, V. Veselovsky, G. Monea, R. West
ACL 2025; ICML 2024 Mechanistic Interpretability Workshop (Spotlight)

nnterp: A Standardized Interface for Mechanistic Interpretability of Transformers
C. Dumas✦
NeurIPS 2025 Mechanistic Interpretability Workshop (Poster)

Diff Mining: Logit Differences Reveal Finetuning Objectives
G. Kocher✦, R. West, **C. Dumas**, J. Minder
ICLR 2026 Workshop on Principled Design for Trustworthy AI

Pre-training Interventions, Ex Post Facto: Grafting Model Beliefs Across Checkpoints
P. Nutter✦, D. Roytburg✦, **C. Dumas**, J. Ou, S. Feng
arXiv 2026

Persona Cartography: Charting Language Model Personality Traits in Weight Space
L. Baines✦, A. G. Hawthorne✦, M. Koroliuk✦, I. Shalibashvili✦, **C. Dumas**, K. Voudouris, D. D. Africa
arXiv 2026

Believe It or Not: How Deeply do LLMs Believe Implanted Facts?
S. Slocum✦, J. Minder, **C. Dumas**, H. Sleight, R. Greenblatt, S. Marks, R. Wang
arXiv 2025

Activation Oracles: Training and Evaluating LLMs as General-Purpose Activation Explainers
A. Karvonen✦, J. Chua, **C. Dumas**, K. Fraser-Taliente, S. Kantamneni, J. Minder, E. Ong, A. Sen Sharma, D. Wen, O. Evans, S. Marks
ICML 2026 (Spotlight)

RESEARCH POSITIONS

Since Apr 2026	**Astra Fellow**, with Owain Evans. Data attribution, then model personas
Sep 2025 - Jan 2026	**SPAR mentor.** Supervised 5 junior researchers on 2 model diffing projects
May 2025 - Apr 2026	**MATS extension**, Neel Nanda's stream. 1-year scholarship to continue my model diffing research
Winter 2025	**MATS 7.0 scholar**, Neel Nanda's stream working on crosscoder and model diffing
Summer 2024	**Research intern, EPFL**, 5 months with Robert West and Chris Wendler
Summer 2023	**Research intern**, 2 months with Jobst Heitzig on Aspiration-Based Q-Learning

EDUCATION

2024-2025	Research Master MVA (Mathematics, Vision, Learning), **ENS Paris-Saclay**
2023-2024	Research Master MPRI, **ENS Paris-Saclay**
2022-2023	Double Bachelor's degree in Computer Science, **ENS Paris-Saclay**
2020-2022	"Classes préparatoires" in Mathematics, Physics and CS, Lycée Thiers

TOOLS AND PROJECTS

tinkerscope	Chat tool for exploring and comparing weird model behaviors across checkpoints
nnterp	A library that standardizes access to LLM internal activations based on NNsight
diffing-toolkit	Implements and visualizes several model diffing techniques
tiny-dashboard	A small app to visualize sparse autoencoder activations
NDIF	Part of the team building the ecosystem behind NNsight
Claude plays Nomic	Multi-agent environment where Claude instances play Nomic, a game where players vote to change the rules as they play
CodinGame	Developed a multiplayer game; top 0.5%, 3% and 7% in bot programming contests (2021-2022)